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Derivatives and the chain rule

Prof. Jane Smith · 2026-10-08 · University of Ljubljana
Recording: 1:31:40

CONTENTS

- 1 The derivative**
 - 1.1 Basic derivatives
- 2 The chain rule**

1 The derivative 03:15

IDEA

The derivative tells you how fast a function changes at a point: the slope of the tangent line.

DEFINITION 1 Derivative

A function f is **differentiable** at a point a if the limit

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

exists.

FOR THE EXAM

"You have to be able to write down the definition of the derivative from memory, it's on every midterm." 07:40

1.1 Basic derivatives 12:05

FORMULA 1 Table of basic derivatives

$f(x)$	$f'(x)$
x^n	nx^{n-1}
e^x	e^x
$\sin x$	$\cos x$

2 The chain rule 34:12

FORMULA 2 Chain rule

$$(f \circ g)'(x) = f'(g(x)) \cdot g'(x)$$

WATCH OUT

Common mistake: forgetting to multiply by the derivative of the inner function.

UNCLEAR IN RECORDING Formula on the board 41:20

The lecturer pointed at the board ("and this cancels out"); the audio doesn't make clear which term.

EXERCISE 1 Derivative of a composite function 52:40

Differentiate $h(x) = \sin(x^2 + 1)$.

work space

SOLUTION

Outer $f(u) = \sin u$, inner $g(x) = x^2 + 1$:

$$h'(x) = \cos(x^2 + 1) \cdot 2x$$

Result: $h'(x) = 2x \cos(x^2 + 1)$

SUMMARY

Concept	Key idea
Derivative	limit of the difference quotient
Chain rule	outer derivative $\times$ inner derivative

Formula sheet

FORMULA 1 Table of basic derivatives

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FORMULA 2 Chain rule

$$(f \circ g)'(x) = f'(g(x)) \cdot g'(x)$$